

HealthConnect Clinic — Week 4 Project Summary

Data Analytics Track | AnalystLab Africa Experience Lab

1. Track Problem

The Data Analytics track's role in the HealthConnect project is to understand the appointment dataset and identify how it can be used to investigate appointment attendance and no-show patterns — defining the business questions and KPIs that will guide deeper analysis in later weeks.

2. Resources Used

- HealthConnect_Appointment_Data.csv — 5,000 fictional appointment records
- HealthConnect_Data_Dictionary.xlsx — variable definitions and data types
- HealthConnect Clinic Knowledge Base — used to cross-check data against stated clinic policy (e.g. opening hours)

3. Key Observations

- The dataset is clean: no duplicate records, no invalid negative values, and no logically impossible field combinations.
- Nearly half of all appointments (48.5%) end in a No-Show, making this a substantial and worthwhile problem to address.
- Booking lead time and prior no-show history are the two strongest observed drivers of no-show risk in this initial pass.
- Reminders show only a modest association with attendance — worth investigating further rather than assuming they are effective.
- One data quality flag was raised: 737 appointments are recorded on Sundays, conflicting with the clinic's stated Sunday closure.

4. Proposed Approach

Five business questions and matching KPIs were defined — covering overall no-show scale, booking lead time, prior no-show history, reminder effectiveness, and distance to clinic — each grounded in a visible pattern in an initial exploratory pass over the data. Full detail is in the accompanying Initial Analysis Document and Notebook.

5. Key Considerations

- All findings so far are correlational, not causal, and based on a single synthetic snapshot of data.
- The Sunday appointment flag needs stakeholder clarification before being treated as either a real pattern or a data artifact.
- Minor missingness in distance and waiting time fields (under 2% each) will need an agreed handling approach before deeper analysis.

6. Proposed Focus for Week 5

- Build out full visualisations for each of the five proposed KPIs.
- Run statistical tests (e.g. chi-square) to confirm which observed associations are statistically meaningful.
- Explore combined/interaction effects, such as lead time and prior no-show history together.
- Quantify lost appointment slots from no-shows versus cancellations to inform a slot-utilisation angle.
- Coordinate with the Data Science track on how the lead-time and prior-history segments identified here could feed into a no-show prediction model.